Discussion 10: 4-vectors in astrophysics – Solutions

Names:

Section:

Now that we've learned how to combine energy and momentum into a 4-vector, which also works for massless particles like photons, we can derive lots of useful relations showing how photon energy (equivalently, frequency or wavelength) transforms with a change in reference frame. In this week's discussion you will derive the general form of the relativistic Doppler effect (first derived by Einstein in his original 1905 paper on relativity) and apply your results to examples in astrophysics, where these techniques are used every day! Throughout, we will practice using natural units with $c = 1$.

Part 1: Relativistic Doppler Effect and Aberration

Consider a light source imoving along the positive x-axis at velocity β with respect to you. The light source emits a photon with energy E_s at an angle θ_s with respect to the x-axis, where both E_s and θ_s are measured in the rest frame of the light source. We are interested in the energy and angle of the photon as you observe them.

1.1. Use the Lorentz transformation of the photon's energy-momentum 4-vector to show that the energy E_r of the photon in your frame is given by

$$E_r = \gamma(1 + \beta \cos\theta_s)E_s. \tag{1}$$

This is the general formula for the relativistic Doppler effect. Check that you recover the longitudinal Doppler effect as derived in class, both for $\theta_s = 0$ (blueshift when the source is moving towards you) and $\theta_s = \pi$ (redshift when the source is moving away from you).

Let the photon's 4-vector in the source frame be $(E_s, E_s\cos\theta_s, E_s\sin\theta_s, 0)$. Performing the Lorentz transformation to take us to the receiver frame,

$$\begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} E_s \\ E_s\cos\theta_s \\ E_s\sin\theta_s \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma E_s(1+\beta\cos\theta_s) \\ \gamma E_s(\beta + \cos\theta_s) \\ E_s\sin\theta_s \\ 0 \end{pmatrix}$$

The zeroth component of this 4-vector is the energy in the receiver frame E_r, so we have immediately

$$E_r = \gamma(1 + \beta\cos\theta_s)E_s.$$

Taking $\theta_s = 0$ gives $E_r = \gamma(1+\beta)E_s$, which we showed in class to be equivalent to $E_r/E_s = \sqrt{\frac{1+\beta}{1-\beta}}$. Taking $\theta_s = \pi$ gives $E_r = \gamma(1-\beta)E_s$, which effectively switches the sign of beta and gives $E_r/E_s = \sqrt{\frac{1-\beta}{1+\beta}}$.

1.2. Show that the angle that the photon makes with the x-axis in *your* frame, θ_r, is given by

$$\tan\theta_r = \frac{\sin\theta_s}{\gamma(\cos\theta_s + \beta)}. \tag{2}$$

We can take the ratio of the x and y components of the boosted 4-vector from problem 1.1 to get $\tan\theta_r$, which immediately gives Eq. (2).

1.3. The fact that $\theta_r \neq \theta_s$ is known as *relativistic aberration*, and makes distant stars in relative motion with respect to us appear that they are oriented differently on the sky. In his original 1905 paper, Einstein wrote this relation as

$$\cos\theta_r = \frac{\cos\theta_s + \beta}{1 + \beta\cos\theta_s} \tag{3}$$

While it's possible to get this relation from Eq. (2) using trig identities, there is a simpler way. Use velocity addition and the fact that $\cos\theta_s$ and $\cos\theta_r$ are the x-components of the light velocity in the source and receiver frame, respectively, to derive Eq. (3).

We perform velocity addition on the x-component of the light velocity only. The x-velocity in the source frame, $\cos\theta_s$ (in units where $c = 1$) is added to the boost velocity β relativistically:

$$v_{x,r} \equiv \cos\theta_r = \frac{\cos\theta_s + \beta}{1 + \beta\cos\theta_s}.$$

1.4. As Morin discusses in Sec. 2.5.2, there are two interpretations of the transverse Doppler effect. Case 1 is when $\theta_s = \pi/2$: the photon is emitted by the source perpendicular to the direction of relative motion, and the source is (instantaneously) at its position of closest approach. Show that taking $\theta_s = \pi/2$ in Eq. (1) recovers Morin eq. (2.39).

With $\theta_s = \pi/2$ we have $\cos\theta_s = 0$, so $E_r = \gamma E_s$; the energy in the receiver's frame is larger.

1.5. Using Eq. (2), find the angle θ_s such that $\theta_r = \pi/2$; in other words, the photon is moving perpendicular to the source velocity in *your* frame. Plug this angle into Eq. (1) to recover Morin's Case 2, eq. (2.40). This is the case where you *see* the source at closest approach, even though (because of aberration) it is actually somewhere else at the instant it emits the light.

When $\theta_r = \pi/2$, $\tan\theta_r$ diverges, which means the denominator vanishes. This gives $\gamma(\cos\theta_s + \beta) = 0$, or $\cos\theta_s = -\beta$. Plugging this in to Eq. (1) gives $E_r = \gamma(1+\beta(-\beta))E_s = \gamma(1-\beta^2)E_s = E_s/\gamma$, so the energy in the receiver's frame is smaller.

Part 2: Relativistic beaming, or the "searchlight effect"

You saw in Part 1 that relative motion can change the angles between the observer and emitter frames. Here we will see some interesting and counterintuitive consequences of that fact. Consider again a source moving at velocity β with respect to you.

2.1. If the source is emitting photons, what is the maximum angle θ_s of emission in the source frame such that the photons travel in the same direction as the source? *Hint: you have effectively solved this in problem 1.5 above.* For highly relativistic speeds, $\beta \approx 1$, and $\theta_s \approx \pi$: if the source is moving towards you, you can see the back of the object as well as its front! This is sometimes known as the "searchlight effect."

For the photons to travel in the same direction as the source in the receiver frame, we need a positive x-component of velocity, so $\cos\theta_r > 0$. Equivalently, $\theta_r < \pi/2$. So the maximum θ_s is exactly what we found in 1.5, $\theta_{s,\max} = \cos^{-1}(-\beta)$. As $\beta \to 1$, we have $\theta_{s,\max} \to \cos^{-1}(-1) = \pi$.

2.2. Suppose the source were stationary; in that case, an observer very far away would only be able to detect photons emitted toward the observer, with $\theta_s \leq \pi/2$. Now let the source move with a boost factor γ. When $\theta_s = \pi/2$ and $\gamma \gg 1$, perform a Taylor expansion of $\tan\theta_r$ to find an approximate value for θ_r. This calculation shows that when the source is highly relativistic, the emitted photons are concentrated into a very small angle. This effect is known as *relativistic beaming*.

With $\theta_s = \pi/2$, we have $\tan\theta_r = \frac{1}{\gamma\beta} \approx \frac{1}{\gamma}$, since when $\gamma \gg 1$, $\beta \approx 1$. Note that we do *not* want to Taylor expand the right-hand side in β, since β is not small! Instead, we use $\tan\theta_r \approx \theta_r + \mathcal{O}(\theta_r^3)$, which is again the small-angle approximation you have probably seen in classical mechanics. This tells us that $\theta_r \approx 1/\gamma \ll 1$.

2.3. Now suppose the source created electrons of mass m_e and momentum p_e at an angle $\theta_s = \pi/2$. Perform a Lorentz transformation on the electron's energy-momentum 4-vector to find the angle θ_r in the receiver frame. If we assume that the electrons are non-relativistic in the source frame ($p_e \ll m_e$), how does your answer compare to problem 2.2? This shows that massive particles are beamed even more than photons.

Doing the same calculation as problem 1.1, this time with a massive 4-vector, gives

$$\begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} E_e \\ p_e\cos\theta_s \\ p_e\sin\theta_s \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma E_e + \gamma\beta p_e\cos\theta_s \\ \gamma\beta E_e + \gamma p_e\cos\theta_s \\ p_e\sin\theta_s \\ 0 \end{pmatrix}$$

The ratio of the y and x components gives us $\tan\theta_r$. Substituting $E_e = \sqrt{p_e^2 + m_e^2}$ and $\theta_s = \pi/2$ gives

$$\theta_r = \tan^{-1}\left(\frac{p_e}{\gamma\beta\sqrt{p_e^2 + m_e^2}}\right).$$

Taking $p_e \ll m_e$, we can Taylor-expand the square root as $\sqrt{p_e^2 + m_e^2} = m_e + \mathcal{O}(p_e^2)$, giving

$$\theta_r \approx \tan^{-1}\left(\frac{p_e}{\gamma\beta m_e}\right)$$

If $\gamma \gg 1$ (and thus $\beta \approx 1$), the receiver-frame angle is smaller by an additional factor $p_e/m_e \ll 1$ compared to the $1/\gamma$ for photons.

Part 3: Velocity measurements with redshift

As mentioned in lecture, quantum mechanics tells us that atoms and molecules emit and absorb light only at specific frequencies. For example, the "H-α" transition of hydrogen occurs in star-forming nebulae as well as the sun's atmosphere, and has a wavelength of 656.277153 nm (you will see why we need so many significant digits in Problem 3.3!). If this line is observed at a slightly different wavelength, one can infer from the Doppler effect that the atoms emitting the light must be in relative motion to the observer.

3.1. The typical velocities of stars in our galaxy are on the order of 300 km/s. Express this velocity in natural units.

With $c \approx 3 \times 10^8$ m/s $= 3 \times 10^5$ km/s, the star's velocity is about $10^{-3}c$, so $v = 10^{-3}$ in natural units.

3.2. Since the velocities from problem 3.1 are small, it is convenient to Taylor-expand the longitudinal Doppler shift formula. Perform this Taylor expansion to leading order in β and write the ratio of wavelengths in the form

$$\frac{\lambda_r}{\lambda_s} = 1 + z \tag{4}$$

where you should find z as a function of β. z is called the *redshift*, since if $z > 0$, the received wavelength is longer (shifted towards the red part of the visible spectrum) than the emitted wavelength.

Defining β such that $\beta > 0$ corresponds to the source moving towards the receiver, we have

$$\frac{\lambda_r}{\lambda_s} = \sqrt{\frac{1-\beta}{1+\beta}} = (1-\beta)^{1/2}(1+\beta)^{-1/2} \approx (1-\beta/2)(1-\beta/2) \approx 1-\beta,$$

so $z = -\beta$. This sign makes sense because if $\beta < 0$, the source is moving away from the receiver, so the frequency is smaller and hence the wavelength is larger (redshifted).

3.3. Redshift can be used to detect exoplanets, or planets orbiting a star outside our solar system. Since the planet and the star are actually orbiting around their mutual center of mass, the star's velocity relative to Earth changes slightly during the course of an orbit. The first exoplanet detected with this "radial velocity" technique was 51 Pegasi b, the discovery of which was awarded the 2019 Nobel Prize. The raw data is shown below: calculate the maximum and minimum wavelengths of the observed H-α line during the course of this planet's orbit. Note the units on the y-axis!

Looking at the central values of the data points, the maximum velocity is about 75 m/s and the minimum is about -110 m/s. (It's worth spending a few moments contemplating how absolutely insane this measurement is. 75 m/s is the speed of a typical train. 30 years ago, we could measure a star 50 light-years away moving away from us with the speed of a train; the current state-of-the-art is 0.45 m/s, which is *walking speed!!*) In natural units, $\beta_{\text{max}} = 2.5 \times 10^{-7}$ and $\beta_{\text{min}} = -3.7 \times 10^{-7}$, so the maximum wavelength is $\lambda_{\text{max}} = (1-\beta_{\text{min}})(656.277153 \text{ nm}) = 656.277396$ nm, and $\lambda_{\text{min}} = (1-\beta_{\text{max}})(656.277153 \text{ nm}) = 656.276989$ nm.

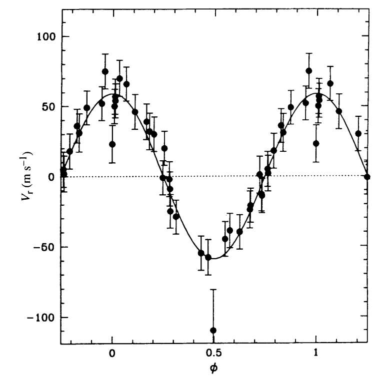

Figure 1: Radial velocity data from 51 Pegasi b discovery, Nature **378**, 355-359 (1995)

3.4. Astronomers in the early 20th century, including Vesto Slipher and Edwin Hubble, found that effectively all distant galaxies were redshifted, implying that they were all moving away from us. Almost 100 years later, the Hubble Space Telescope observed "deep field" galaxies that have a redshift of $z = 12$. If we take Eq. (4) at face value, what is the velocity of these galaxies? You should find that it is greater than 1 in natural units; have we discovered a violation of special relativity with an object moving faster than the speed of light?! No, but the answer is even more amazing: Eq. (4) does not apply on very large distance scales because *the universe itself is expanding.* The redshifts of distant galaxies were the first pieces of evidence for an expanding universe!

Taking $z = -\beta$, the galaxies appear to be receding at a velocity 12 times the speed of light!